16.04.26

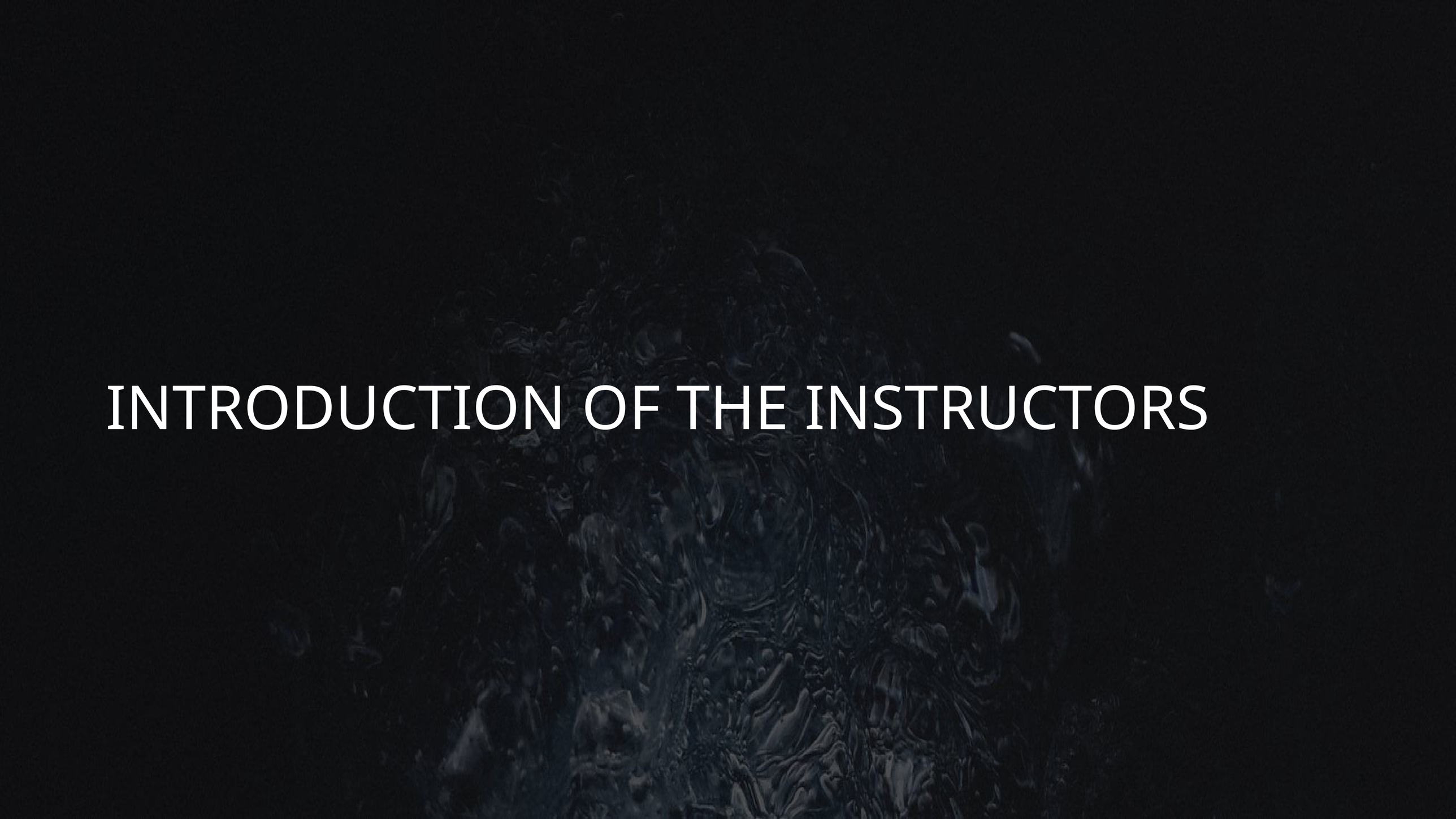
OPENCAMPUS.sh

Applied Machine Learning

GENERAL INTRODUCTION

- Introduction of the Instructors
- Intro to opencampus.sh
- Organizational Matters
- Introductory Discussion on AI
- Course Projects
- ML Frameworks

INTRODUCTION OF THE INSTRUCTORS



OPENCAMPUS.sh

- Nonprofit organization which oversees a variety of initiatives
- Offering a wide range of educational opportunities, support, and networking for entrepreneurs, creatives, and anyone curious, regardless of age, educational background, or origin
- The services are open to everyone and mostly free.
- The goal is to support the entrepreneurial landscape, promote creative change processes, and contribute to innovative and sustainable future development.









starter kitchen.de



COBL

COZY WORKING, CULTURE & EVENTS





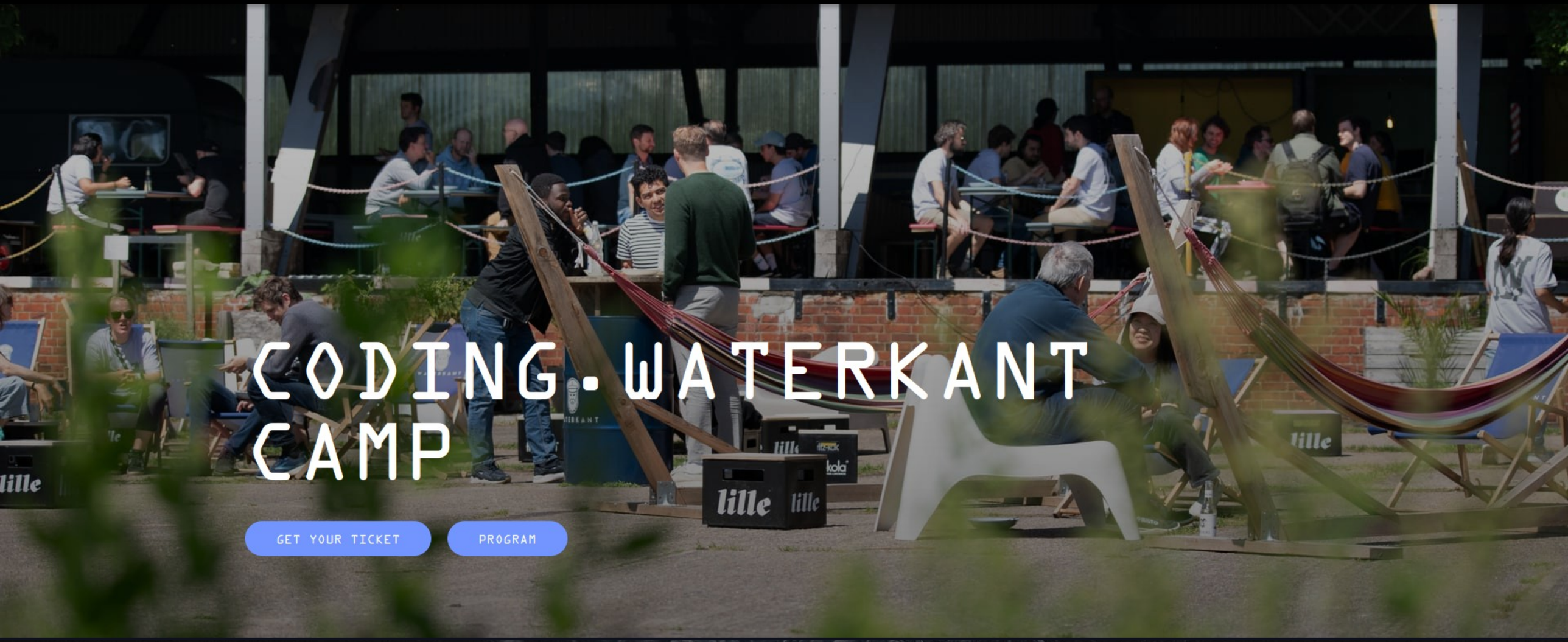


FABLAB KIEL


Visual Science
Communication
Research
WAS SIND OZEANE WERT?
HOW MUCH ARE OCEANS WORTH?
WATERKANT
FESTIVAL
June 18-19

CW

CODING CAMP EDUCATION COMMUNITY ABOUT US



CODING.WATERKANT CAMP

GET YOUR TICKET PROGRAM

CW

CODING CAMP EDUCATION COMMUNITY ABOUT US

Coding.Waterkant Festival is a hands-on, community-driven tech camp that brings together developers, ML specialists, students, and professionals to collaborate, create, and learn.

Over several days, participants dive into coding projects, explore cutting-edge technologies, and exchange ideas in a collaborative, open environment by the Baltic Sea.

Whether you're part of a startup, a tech company, or an independent developer, you can work on your own project or join a team. The focus is on practical learning, real-world solutions, and fostering lasting collaborations.

GET YOUR TICKET PROGRAM

INTENSIVE CODING
on individual or collaborative projects

HANDS-ON WORKSHOPS
in Machine Learning, Open Source, and Robotics

INSPIRATION, EXCHANGE, AND NETWORKING
at eye level

ON-SITE CATERING & ACCOMMODATION
to fully immerse yourself in the experience

https://coding.waterkant.sh





Coding.Waterkant

4.6 • 141 Bewertungen

Kiel, Deutschland

1.141 Mitglieder

Teilen

Info | Events | Mitglieder | Fotos | Diskussionen | Mehr

Gruppe verwalten | Event erstellen

Über uns

Our meetup is organized with support of opencampus.sh and the Digitale Wirtschaft Schleswig-Holstein (DiWiSH).

It is for hobbyists, academics as well as professionals who are interested in the latest developments in AI, Data Science, and Machine Learning as well as interesting applications of these technologies.

Our focus is more on the technology side, but everyone who is interested in this topic is more than welcome. We are explicitly not commercial and organize this meetup because we are passionate about AI, not to promote some product or service.

Organisatoren

opencampus.sh und 9 andere

Nachricht

opencampus.sh ist ein **Super-Organisator**

Mitglieder 1.141

Alle anzeigen

ORGANIZATIONAL MATTERS

Attendance Registration:

Online

- **Use your full names in the zoom meetings!**
- **Turn your camera on!**
- **Take part via laptop or desktop in a sufficiently quite environment!**

Presence

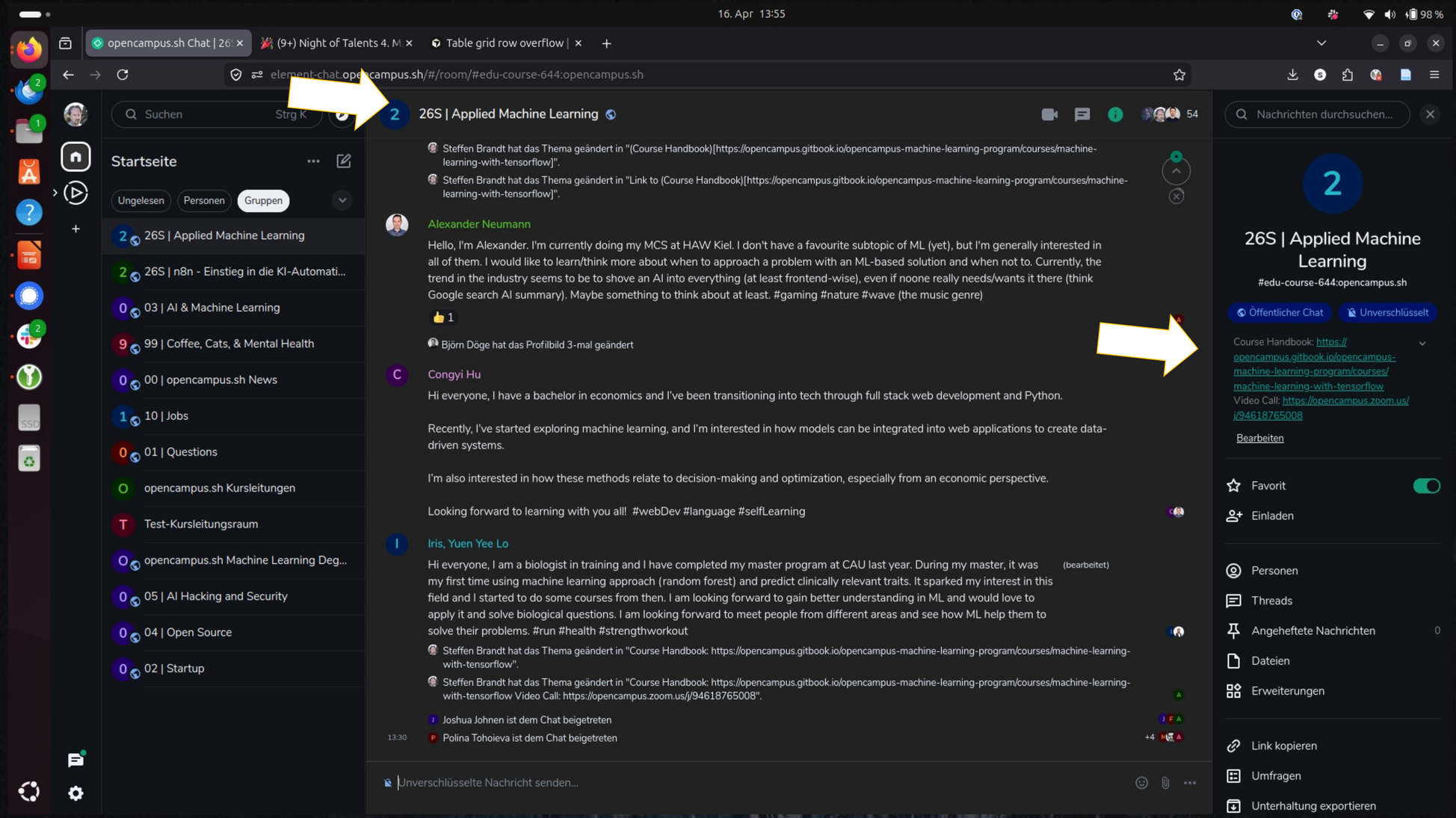
- Scan the QR-Code if you participate in presence

Please write us if you will not go on with the course!

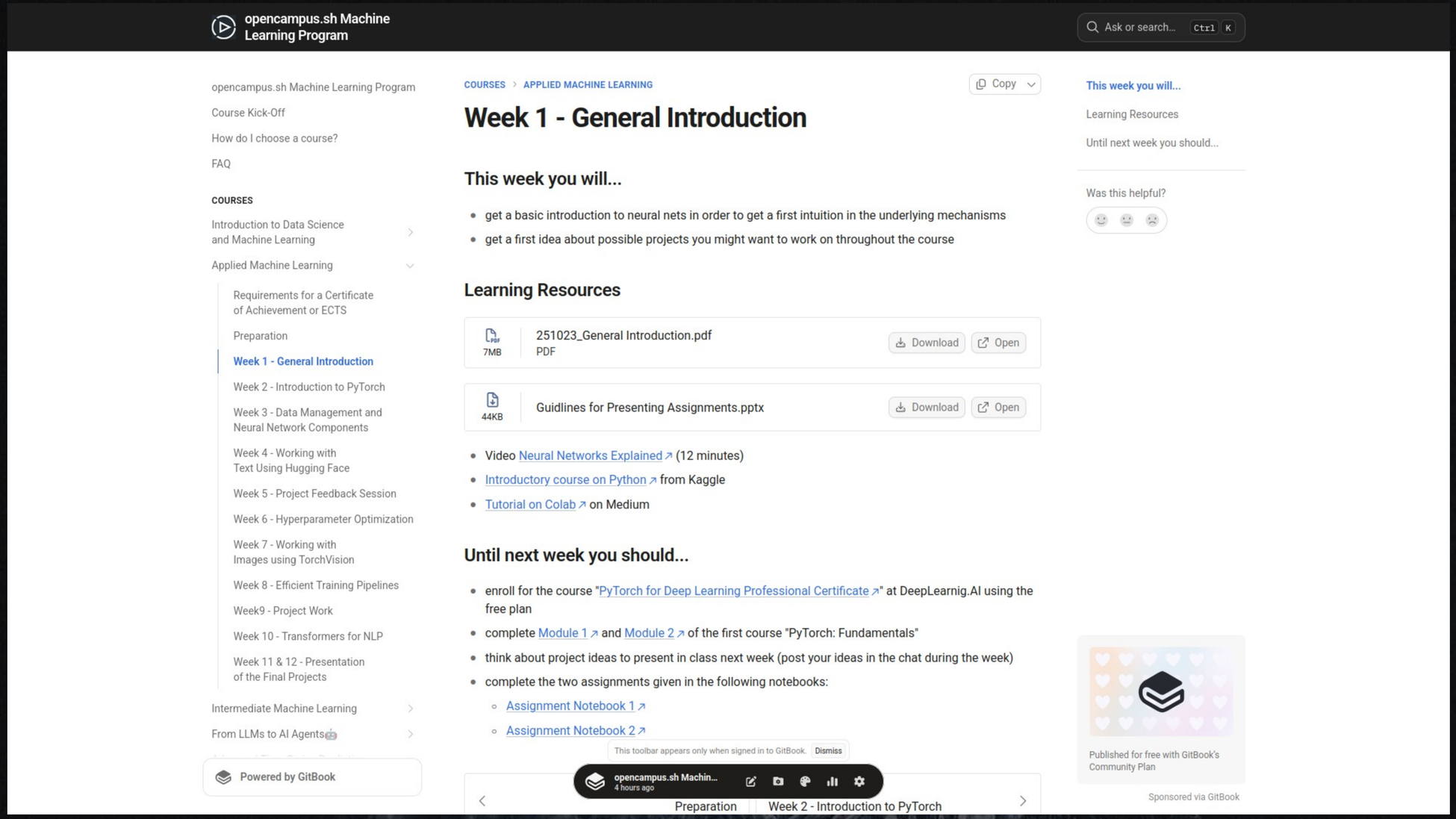
CHAT

- Please, always ask any questions to us in the chat of the course instead of using private messages.
- For questions that are not considering the course project or content, for example on ECTS recognition use the opencampus questions chat or write at edu@opencampus.sh.

ELEMENT CHAT



COURSE HANDBOOK



ZOOM

- Try the different viewing modes:
  - Gallery View/ Active Speaker
  - Split Screen/ Full Screen Mode

- Maybe watch this video to get an idea: https://www.youtube.com/watch?v=v3IPAbpVjd4

16.04.2026 General Introduction
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE



Steffen Brandt
Referent:in
LinkedIn-Profil

23.04.2026 Introduction into PyTorch
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE



Carl Schmidt
Referent:in
LinkedIn-Profil

30.04.2026 Data Management and Neural Network Components
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE



Carl Schmidt
Referent:in
LinkedIn-Profil

07.05.2026 Working with Text using Hugging Face
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE



Jake Petersen
Referent:in

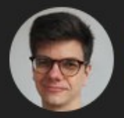
14.05.2026 Feedback Sessions
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE

21.05.2026 Hyperparameter Optimization
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE



Hannes Körner
Referent:in

28.05.2026 Working with Images using TorchVision
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE



Hannes Körner
Referent:in

04.06.2026 Efficient Training Pipelines
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE



Steffen Brandt
Referent:in
LinkedIn-Profil

11.06.2026 Project Work
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE

18.06.2026 Transformers for NLP in PyTorch
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE



Jake Petersen
Referent:in

25.06.2026 Project Presentations - Part 1
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE



Steffen Brandt
Referent:in
LinkedIn-Profil

02.07.2026 Project Presentations - Part 2
18:00 - 19:30
Gebäude 2.4, Fritz-Thiedemann-Ring 20, Heide + Kuhnkestr. 6, 24118 Kiel + ONLINE

PROJECT MILESTONES

23.04.	Present your Ideas
30.04.	Form Groups
07.05.	Literature Review
14.05.	Dataset Characteristics and Individual Feedback Sessions

Deadline for completing the repo sections: 17.05.

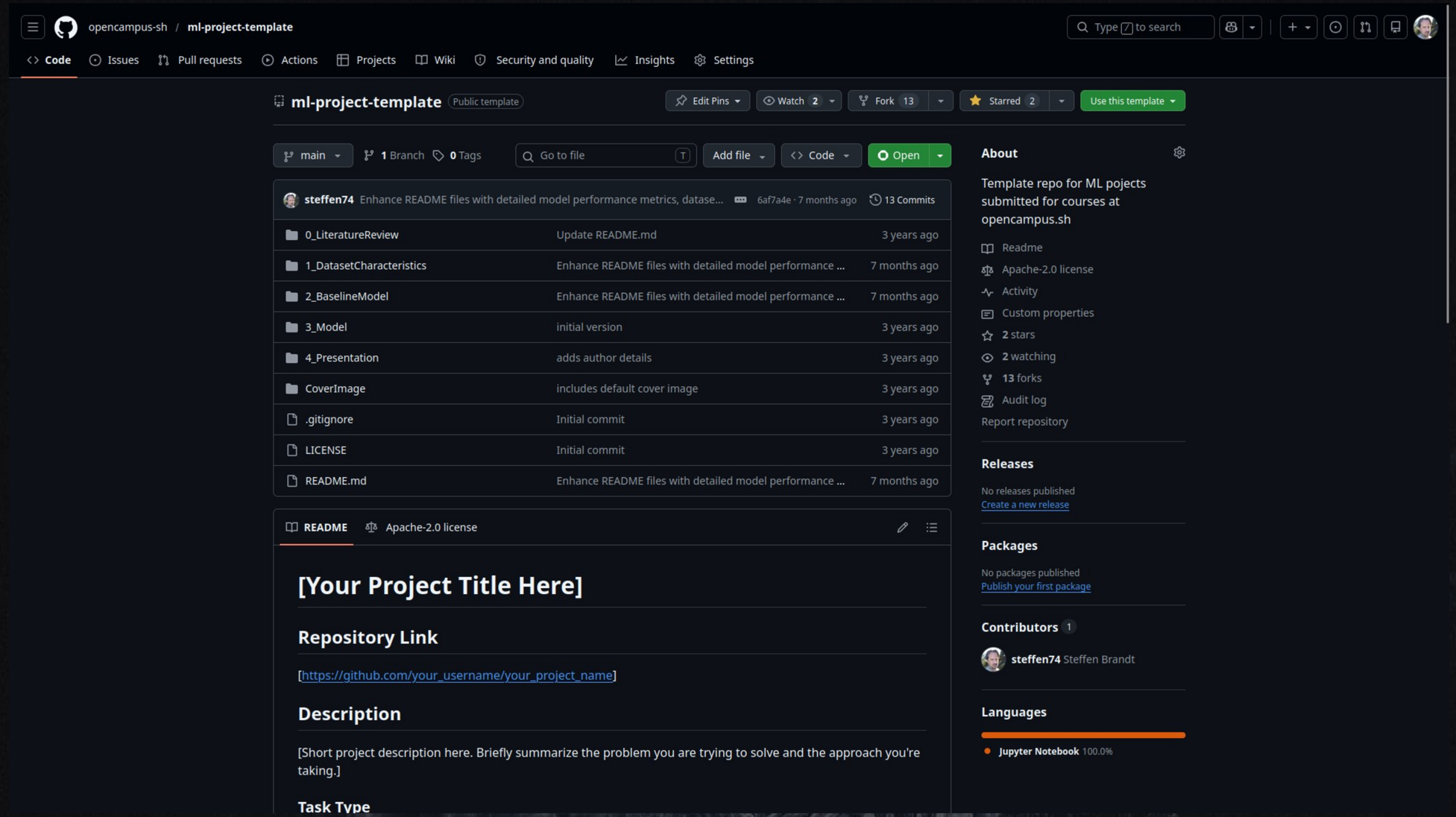
21.05.	Definition of Model Evaluation
28.05.	Project Work
04.06.	Baseline Model Estimation

Deadline for completing the repo sections: 07.06.

11.06.	Project Work / Optional Feedback Sessions (*no regular class in this week!*)
25.06.	Project Presentations, Part I
02.07.	Project Presentations, Part II

Submission deadline for the documented repo: 01.08.2026

TEMPLATE REPO



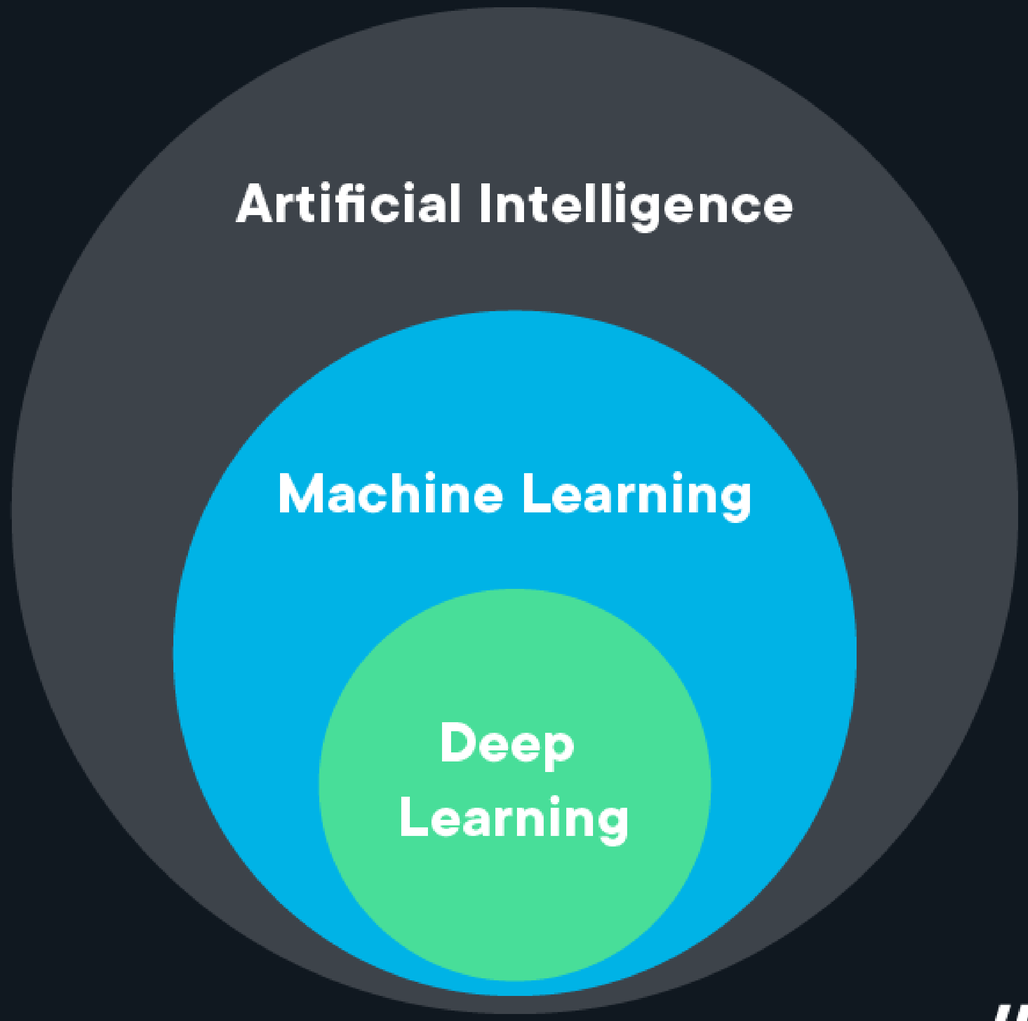
FIRST BREAKOUT

- 15-20 Minutes
- Present yourself
- Discussion Questions:
  - Do you know examples for Machine Learning?
  - Do you know examples for Deep Learning?

Artificial Intelligence
A science devoted to making machines think and act like humans.

Machine Learning
Focuses on enabling computers to perform tasks without explicit programming.

Deep Learning
A subset of machine learning based on artificial neural networks.

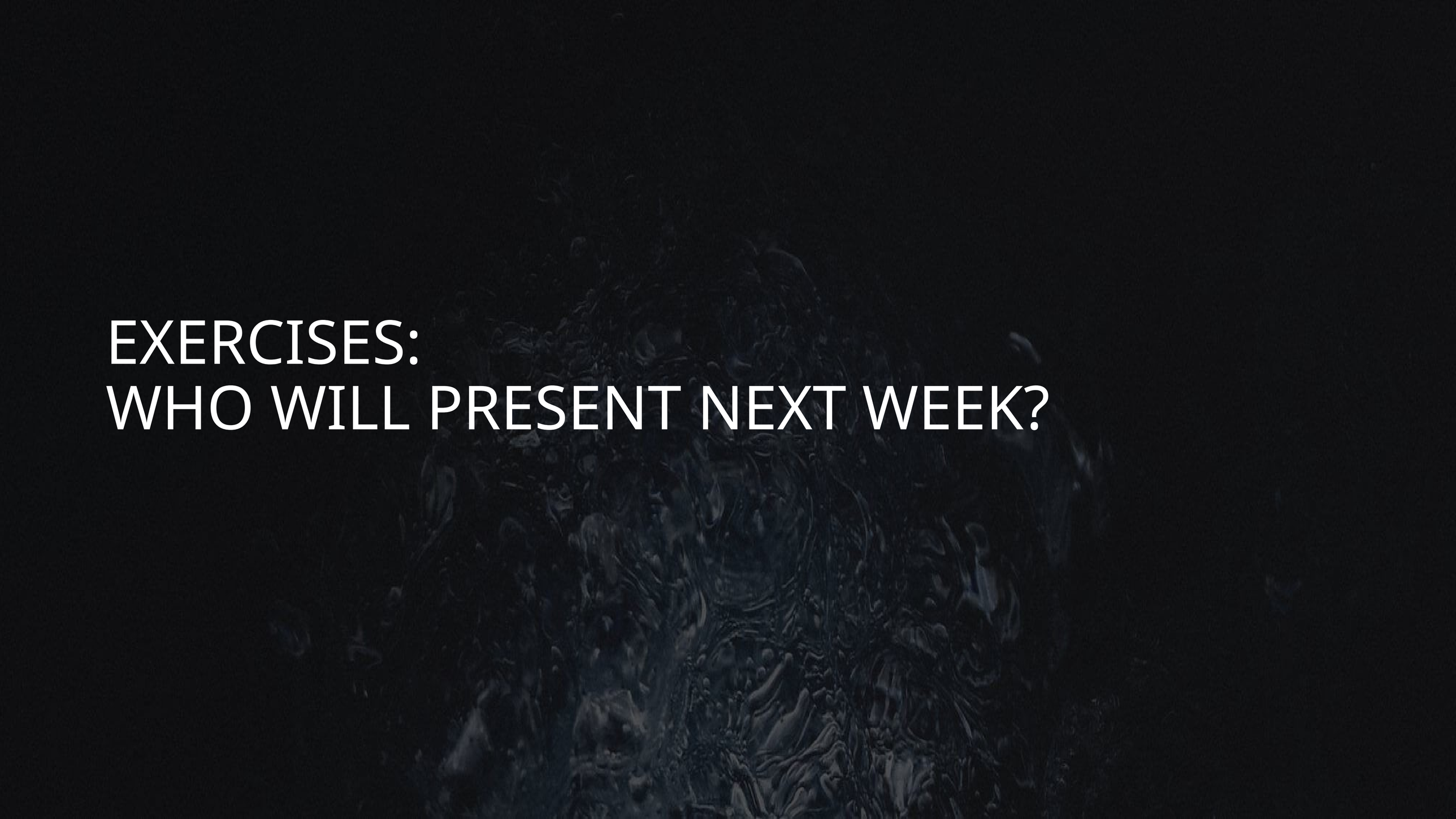


Deep Learning vs. Machine Learning—What's the Difference? (2021, February 8). Flatiron School. https://flatironschool.com/blog/deep-learning-vs-machine-lea

EXERCISES

- Each week two to four of you will present the exercises given in the course handbook
- Check the guidelines to present the exercise given in the course handbook!
- Each of you presents at least once

EXERCISES:
WHO WILL PRESENT NEXT WEEK?



PROJECTS

- Bring your own idea and data
- Choose a project from the Challenge Data platform

ChallengeData By MathA

HOME CHALLENGES FAQ CONTACT LOGIN EN



AGIR POUR L'ÉDUCATION
UN ENJEU SCIENTIFIQUE POUR LA SOCIÉTÉ

Welcome to the Challenge Data website!!

Each year, we organize machine learning challenges from data provided by public services, companies and laboratories: **general documentation** and **FAQ**. Seasons begin in January; the challenges are introduced in the context of Stéphane Mallat's lesson at the Collège de France.

A prize ceremony for the best participants of the preceding season will be held in February at the College de France (03/02/2022).

For participants

Guide to create an account, choose your challenges and submit solutions.

RELEVANCE OF THE PROJECTS

- Most important for a career in ML will be work experience and your GitHub profile
- Focus on building a noteworthy GitHub project repository
- Use the template repository
- Outstanding projects will be nominated for the VDE Machine Learning Prize, which is awarded once a year

EVENTS

COURSE PROJECTS

Choosing a Project

How to Start, Complete, and Submit Your Project

ADDITIONAL RESOURSES

Glossary

Coursera

Selecting the Optimizer

Choosing the Learning Rate

Learning Linear Algebra

Learning Python

Support Vector Machines

ML Statistics

TOOLS

Git

RStudio

Google Colab

Powered by GitBook

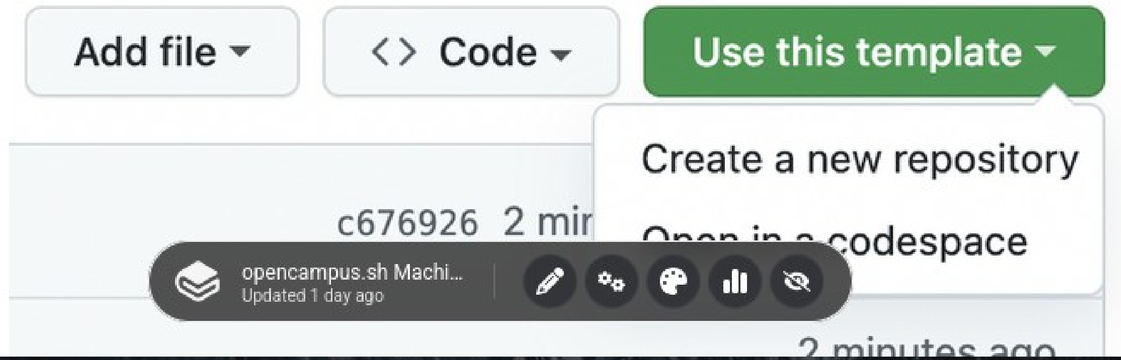
How to Start, Complete, and Submit Your Project

In all Machine Learning courses you have:

- to complete a machine learning project in a team of up to 4 participants,
- attend at least all but 2 sessions of the course, and
- use the provided project template repository for documentation (unless otherwise instructed).

Starting Your Project

1. **Navigate to the** Template Repository ↗
2. **Use this Template**: Above the file list, click the "Use this template" button.



Starting Your Project

Working on Your Project

Submitting Your Project

Was this helpful?

Published for free with GitBook's Community Plan

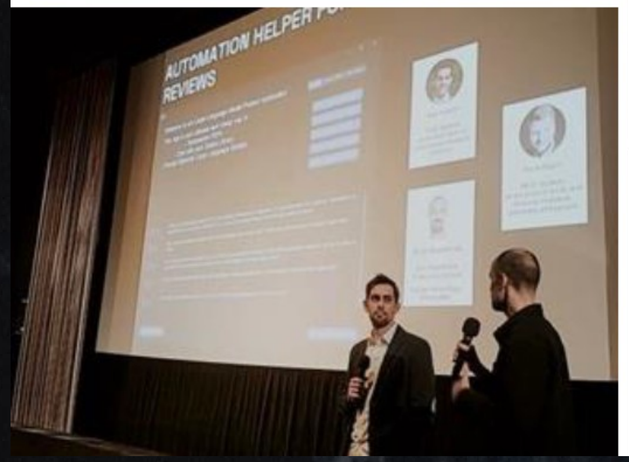
Sponsored via GitBook

VDE

SPECIAL PRIZE MACHINE LEARNING













COURSES

Fine-Tuning and Deployment of Large Language Models

Archive

EVENTS

Coding.Waterkant 2023

Prototyping Week

PROJECTS

How to Start, Complete, and Submit Your Project

Possible Projects

Past Projects

ADDITIONAL RESOURSES

Glossary

Coursera

Selecting the Optimizer

Choosing the Learning Rate

Learning Linear Algebra

Learning Python

Support Vector Machines

ML Statistics

TOOLS

Git

RStudio

Google Colab

Zoom

Powered by GitBook

How to Start, Complete, and Submit Your Project

Starting Your Project

Working on Your Project

Submitting Your Project

Was this helpful?

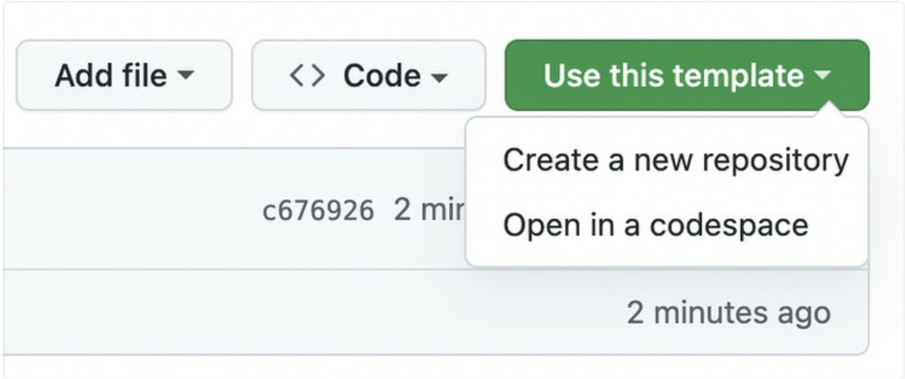
Export as PDF

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Starting Your Project

1. **Navigate to the** Template Repository
2. **Use this Template**: Above the file list, click the "Use this template" button.



Use this template button

3. **Create Repository from Template**: You'll be prompted to name your new repository and you can choose whether it should be public or private. You'll also have the option to include all branches in the template repository, if there are more than one.
4. **Create Repository**: Click "Create repository from template" to create the new repository.
5. **Clone the New Repository**: You can now clone the new repository to your local machine using `git clone` and start working on your project.

Working on Your Project

FIRST PROJECT IDEAS



ML LIBRARIES (LOWER LEVEL)







PYTORCH

ML LIBRARIES (HIGHER LEVEL)



Transformers

spaCy



haystack
by deepset



LangChain

LUDWIG



Auto
gpt

DEVELOPMENT ENVIRONMENTS













TASKS UNTIL NEXT WEEK

- Completion of the course material linked in the course handbook
- Write about your project interest in the course chat
- Complete the two assignments given in the following notebooks:
  - Assignment Notebook
- Bring questions!